

FLORENT TARIOLLE

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Engineering student seeking a 6-month final-year research internship in deep learning, starting in April 2027.

EDUCATION

INSA Rouen Normandy	Rouen, France
Engineering degree, Computer Science and Artificial Intelligence, expected 2027	2023 – 2027
<i>Approx. U.S.-equivalent GPA: 3.7/4.0 overall; 4.0/4.0 latest semester.</i>	
University of Rouen Normandy	Rouen, France
Master's Degree, Machine Learning for Artificial Intelligence, expected 2027	2026 – 2027
Lycée Dupuy de Lôme	Lorient, France
CPGE MP (Preparatory Classes for Grandes Écoles, Mathematics and Physics)	2021 - 2023

PUBLICATIONS

Opportunistic Target Selection	May 2026
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F. Tariolle and F. Yger. *CAp 2026 (Conférence sur l'Apprentissage automatique)*.

- Introduced a lightweight wrapper for score-based black-box adversarial attacks using early class confidence signals to reduce class drift in undirected random-search attacks.
- Improved attacks by up to 27 percentage points in success rate and a 43% relative reduction in censored-mean iterations on ResNet-50, validated across 4,500 ImageNet attack runs on five standard classifiers.

PATENTS

Neural Network for Adaptive Bilateral Filtering	Filed Jul 2026
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F. Lefebvre, F. Tariolle, F. Schnitzler, H. Guermoud, and S. Yassin. *Patent application*.

- Neural-network adaptive bilateral filtering (ABIF) for reconstructed video blocks inside the in-loop filtering process: feature-map extraction, learned filter-profile selection, and encoder rate-distortion-driven configuration signaling for decoder-side filtering.

EXPERIENCE

InterDigital	Rennes, France
<i>Deep Learning Research Intern</i>	May 2026 - Present

- Working on deep learning methods for next-generation video compression, focusing on VVC (H.266) in-loop filtering and decoder-side filter selection to improve reconstructed video quality and coding efficiency.
- Co-invented a neural-network adaptive bilateral filtering method for in-loop reconstruction; patent application filed in July 2026.

Enedis	Rouen, France
<i>Machine Learning Intern</i>	Jan 2026 - May 2026

- Working on deep generative models and large-scale pipelines to simulate high-frequency time series for 38 million customers, supporting a digital twin of the French electricity grid.

RESEARCH PROJECTS

DashVMC: Real-Time Discrete World Model Control in Geometry Dash	Feb 2026 - Present
<i>Independent research project; manuscript in preparation.</i>	

- Built a real-time Geometry Dash controller around an 8x8 FSQ token grid, action-conditioned transformer dynamics, and a lightweight actor-critic initialized by behavioural cloning and refined with PPO in frozen-model rollouts.
- Trained offline from approximately two hours of recorded gameplay with no live-game interaction; deployed a decoder-free 15.57M-parameter controller at 60 FPS on an RTX 2060, improving over its behavioural-cloning parent on all three official levels and a held-out community layout.

AWARDS

Finalist, Hack the World(s) Hackathon	Jun 2026
<i>24-hour team project at a Yann LeCun-sponsored hackathon on JEPAs and world models</i>	

- Developed a geometry-aware joint-embedding EEG SSL system: strong SIGReg frozen-transfer baselines and Riemannian latent analysis showing structure in SPD covariance representations.

OPEN SOURCE

Rose (25K+ DAU; 400+ GitHub stars)	Sep 2025 - Present
<i>Co-founder & Lead Developer</i>	

- Co-founded and led a real-time customization tool used by 25K+ daily active users; coordinated a 6-person team and designed its Python backend, Cloudflare Workers WebSocket relay, and browser JavaScript integration.
- Replaced OCR-based state extraction with deterministic DOM parsing, reducing latency and complexity.

VisualTorch (300+ GitHub stars)	Jul 2026 - Present
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Maintainer & MCP Integration Lead

- Maintain VisualTorch, an official PyTorch Ecosystem project; designed and implemented its MCP integration for generating editable neural-network architecture diagrams.

TECHNICAL SKILLS

Programming: Python, C++

AI & ML: PyTorch, CUDA; Deep Learning, Self-Supervised Learning, World Models, Model-Based RL and Planning

Languages: French (native), English (TOEIC 970/990)

REFERENCES

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Frédéric Lefèbvre: frederic.lefebvre@interdigital.com